

		F on wire Y will be doubled since current Y is doubled.
24	A	Area of coil $A = \pi (0.16 / 2)^2 = 0.0201 \text{ m}^2$ Induced emf $V = \Delta\phi / \Delta t$ $= A \Delta B / \Delta t$ $= 0.0201 \times (5.00 - 0) / 10.0 \times 10^{-3}$ $= 10.053 \text{ V}$ Current $= V / R = 10.053 / 4.0 = 2.51 \text{ A}$ To oppose change in B field, coil will induce B field into the page. Using right hand grip rule, induced current in clockwise direction.
25	A	Reducing speed by half will decrease emf by half, and hence the coil experiences